Full Mock Test 1 · Paper 1

Question 1

A square has centre $(3, 4)$ and one corner at $(1, 5)$. Another corner is at

A $(1, 3)$

B $(5, 5)$

C $(2, 2)$

D $(5, 2)$

E $(4, 2)$

Full Mock Test 1 · Paper 1

Question 2

What is the value of $\int_0^1 (e^x - x)(e^x + x)\,dx$?

A $\dfrac{3e^2 - 5}{6}$

B $\dfrac{e^2 + 3}{6}$

C $\dfrac{3e^2 - 2}{6}$

D $\dfrac{2e^2 - 3}{6}$

E $\dfrac{3e^2 + 2}{6}$

Full Mock Test 1 · Paper 1

Question 3

A regular dodecagon is a 12-sided polygon with all sides the same length and all internal angles equal. If I construct a regular dodecagon by connecting 12 equally-spaced points on a circle of radius 1, then the area of this polygon is

A 3

B $3\sqrt{3}$

C $3\sqrt{2}$

D $2\sqrt{2}$

E $6+3\sqrt{3}$

Full Mock Test 1 · Paper 1

Question 4

A particle moves on a straight line with velocity $v(t) = k\sqrt{t} - t$ metres per second for $0 \le t \le 9$. Its displacement during the first four seconds is 8 metres. How far does the particle travel during the full nine seconds?

A 9 m

B 12 m

C $\dfrac{27}{2}$ m

D 18 m

E $\dfrac{45}{2}$ m

Full Mock Test 1 · Paper 1

Question 5

Tangents to $y = e^x$ are drawn at (p, e^p) and (q, e^q). These tangents cross the x-axis at a and b respectively. It follows that, for all p and q,

A $pa = qb$

B $p - a = q - b$

C $p - a < q - b$

D $p + q = a + b$

E $p - a > q - b$

Full Mock Test 1 · Paper 1

Question 6

How many real solutions x are there to the equation $x|x| + 1 = 3|x|$?
Note that $|x|$ is equal to x if $x \geq 0$, and equal to $-x$ otherwise.

A 1

B 3

C 4

D 0

E 2

Full Mock Test 1 · Paper 1

Question 7

If the expression $\left(x+1+\frac{1}{x}\right)^4$ is fully expanded term-by-term and like terms are collected together, there is one term which is independent of x. The value of this term is

A 51

B 81

C 10

D 19

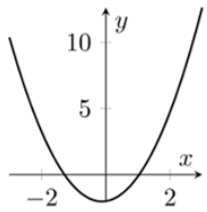
E 14

Full Mock Test 1 · Paper 1

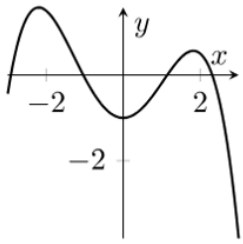
Question 8

The following five graphs are, in some order, plots of $y = f(x)$, $y = g(x)$, $y = h(x)$, $y = \frac{df}{dx}$ and $y = \frac{dg}{dx}$; that is, three unknown functions and the derivatives of the first two of those functions. Which graph is a plot of $h(x)$?

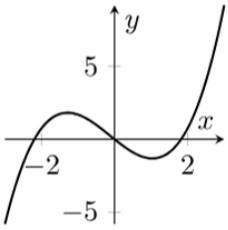
A



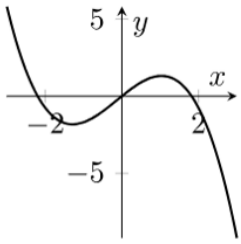
B



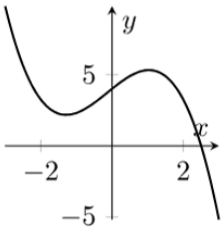
C



D



E



Full Mock Test 1 · Paper 1

Question 9

Let s_k denote the sum of the kth powers of the roots of the polynomial $x^3 - 5x^2 + 8x - 13$. In particular, $s_0 = 3$, $s_1 = 5$, and $s_2 = 9$. Let $a, b,$ and c be real numbers such that $s_{k+1} = as_k + bs_{k-1} + cs_{k-2}$ for $k = 2, 3, \ldots$. What is $a + b + c$?

A -6

B 0

C 6

D 10

E 26

Full Mock Test 1 · Paper 1

Question 10

In $\triangle ABC$ with integer side lengths, $\cos A = \frac{11}{16}$, $\cos B = \frac{7}{8}$, and $\cos C = -\frac{1}{4}$. What is the least possible perimeter for $\triangle ABC$?

A 9

B 12

C 23

D 27

E 44

Full Mock Test 1 · Paper 1

Question 11

Define binary operations $\diamond$ and $\heartsuit$ by

$$a \diamond b = a^{\log_7(b)}, \qquad a \heartsuit b = a^{\frac{1}{\log_7(b)}}.$$

for all real numbers a and b for which these expressions are defined.

The sequence (a_n) is defined recursively for $n \geq 4$ by

$$a_3 = 3 \heartsuit 2$$
$$a_n = \big(n \heartsuit (n-1)\big) \diamond a_{n-1}$$

To the nearest integer, what is $\log_7(a_{2019})$?

A 8

B 9

C 10

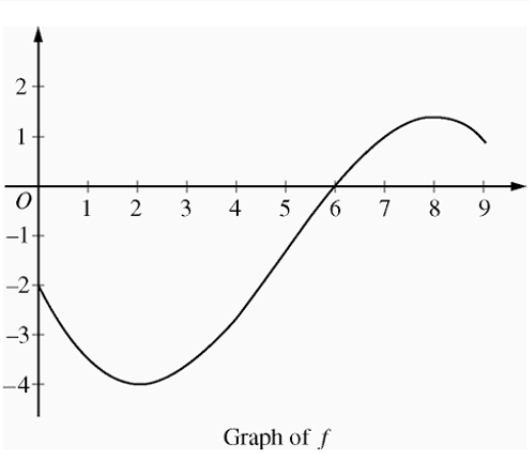
D 11

E 12

Full Mock Test 1 · Paper 1

Question 12

The graph of a differentiable function f is shown below. If $h(x) = \int_0^x f(t)\,dt$, which of the following is true?



Graph of f

A $h(6) < h'(6) < h''(6)$

B $h(6) < h''(6) < h'(6)$

C $h'(6) < h(6) < h''(6)$

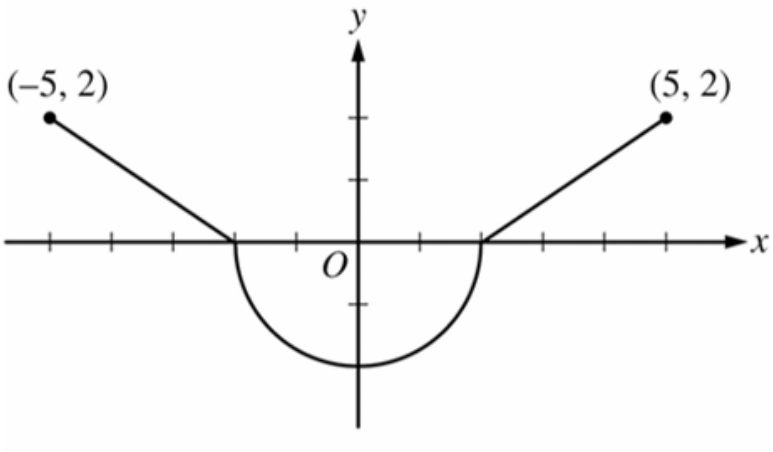
D $h''(6) < h(6) < h'(6)$

E $h''(6) < h'(6) < h(6)$

Full Mock Test 1 · Paper 1

Question 13

The graph of f', the derivative of a function f, consists of two line segments and a semicircle, as shown below. If $f(2) = 1$, then $f(-5) =$



A $2\pi - 2$

B $2\pi - 3$

C $2\pi - 5$

D $6 - 2\pi$

E $4 - 2\pi$

Full Mock Test 1 · Paper 1

Question 14

The function f is defined by $f(x) = \frac{x}{x+2}$. What points (x, y) on the graph of f have the property that the line tangent to f at (x, y) has slope $\frac{1}{2}$?

A $(0, 0)$ only

B $\left(\frac{1}{2}, \frac{1}{5}\right)$ only

C $(0, 0)$ and $(-4, 2)$

D $(0, 0)$ and $\left(4, \frac{2}{3}\right)$

E There are no such points.

Full Mock Test 1 · Paper 1

Question 15

A sequence of numbers is defined recursively by $a_1 = 1$, $a_2 = \frac{3}{7}$, and

$$a_n = \frac{a_{n-2} \cdot a_{n-1}}{2a_{n-2} - a_{n-1}} \quad \text{for all } n \geq 3.$$

Then a_{2019} can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?

A 2020

B 4039

C 6057

D 6061

E 8078

Full Mock Test 1 · Paper 1

Question 16

Let g be a function with first derivative given by $g'(x) = \int_0^x e^{-t^2}\,dt$. Which of the following must be true on the interval $0 < x < 2$?

A g is increasing, and the graph of g is concave up.

B g is increasing, and the graph of g is concave down.

C g is decreasing, and the graph of g is concave up.

D g is decreasing, and the graph of g is concave down.

E g is decreasing, and the graph of g has a point of inflection on $0 < x < 2$.

Full Mock Test 1 · Paper 1

Question 17

Positive real numbers $x \neq 1$ and $y \neq 1$ satisfy $\log_2 x = \log_y 16$ and $xy = 64$. What is $\left(\log_2 \frac{x}{y}\right)^2$?

A $\frac{25}{2}$

B 20

C $\frac{45}{2}$

D 25

E 32

Full Mock Test 1 · Paper 1

Question 18

How many solutions does the equation $\tan(2x) = \cos\left(\frac{x}{2}\right)$ have on the interval $[0, 2\pi]$?

A 1

B 2

C 3

D 4

E 5

Full Mock Test 1 · Paper 1

Question 19

What is the sum of all possible values of k for which the polynomials $x^2 - 3x + 2$ and $x^2 - 5x + k$ have a root in common?

A 3

B 4

C 5

D 6

E 10

Full Mock Test 1 · Paper 1

Question 20

An even number of circles are nested, starting with a radius of 1 and increasing by 1 each time, all sharing a common point. The region between every other circle is shaded, starting with the region inside the circle of radius 2 but outside the circle of radius 1. An example showing 8 circles is displayed below. What is the least number of circles needed to make the total shaded area at least 2023π?



A 46

B 48

C 56

D 60

E 64

Full Mock Test 1 · Paper 2

Question 1

A number is chosen at random from among the first 100 positive integers, and a positive integer divisor of that number is then chosen at random. What is the probability that the chosen divisor is divisible by 11?

A $\frac{4}{100}$

B $\frac{9}{200}$

C $\frac{1}{20}$

D $\frac{11}{200}$

E $\frac{3}{50}$

Full Mock Test 1 · Paper 2

Question 2

Let $P(x)$ be the unique polynomial of minimal degree with the following properties:
• $P(x)$ has a leading coefficient 1,
• 1 is a root of $P(x) - 1$,
• 2 is a root of $P(x - 2)$,
• 3 is a root of $P(3x)$, and
• 4 is a root of $4P(x)$.
The roots of $P(x)$ are integers, with one exception. The root that is not an integer can be written as $\frac{m}{n}$, where m and n are relatively prime integers. What is $m + n$?

A 41

B 43

C 45

D 47

E 49

Full Mock Test 1 · Paper 2

Question 3

How many three-digit positive integers N satisfy the following properties?

- N is divisible by 7.
- The number formed by reversing the digits of N is divisible by 5.

A 13

B 14

C 15

D 16

E 17

Full Mock Test 1 · Paper 2

Question 4

Let $f(x) = 3 - 2x$. A differentiable function g satisfies $g'(x) = f(x)f'(x)(x - 3)$.
For this question, take "g is increasing at x" to mean $g'(x) \geq 0$.
Decide which of the following statements are true.

1. $x \leq \frac{3}{2}$ is a *sufficient* condition for g to be increasing at x.
2. $x \geq 3$ is a *necessary* condition for g to be increasing at x.
3. $x \leq \frac{3}{2}$ or $x \geq 3$ is a *necessary and sufficient* condition for g to be increasing at x.

How many of the statements 1–3 are true?

A 0

B 1

C 2

D 3

Full Mock Test 1 · Paper 2

Question 5

Maureen is keeping track of the mean of her quiz scores this semester. If Maureen scores an 11 on the next quiz, her mean will increase by 1. If she scores an 11 on each of the next three quizzes, her mean will increase by 2. What is the mean of her quiz scores currently?

A 4

B 5

C 6

D 7

E 8

Full Mock Test 1 · Paper 2

Question 6

What is the least value of n such that $n!$ is a multiple of 2024?

A 11

B 21

C 22

D 23

E 253

Full Mock Test 1 · Paper 2

Question 7

How many angles θ with $0 \leq \theta \leq 2\pi$ satisfy $\log(\sin(3\theta)) + \log(\cos(2\theta)) = 0$?

A 0

B 1

C 2

D 3

E 4

Full Mock Test 1 · Paper 2

Question 8

On Halloween 31 children walked into the principal's office asking for candy. They can be classified into three types: some always lie; some always tell the truth; and some alternately lie and tell the truth. The alternaters arbitrarily choose their first response, either a lie or the truth, but each subsequent statement has the opposite truth value from its predecessor. The principal asked everyone the same three questions in this order.

"Are you a truth-teller?" The principal gave a piece of candy to each of the 22 children who answered yes.
"Are you an alternater?" The principal gave a piece of candy to each of the 15 children who answered yes.
"Are you a liar?" The principal gave a piece of candy to each of the 9 children who answered yes.

How many pieces of candy in all did the principal give to the children who always tell the truth?

A 7

B 12

C 21

D 27

E 31

Full Mock Test 1 · Paper 2

Question 9

Which expression is equal to $\left|a-2-\sqrt{(a-1)^2}\right|$ for $a<0$?

A $3-2a$

B $1-a$

C 1

D $a+1$

E 3

Full Mock Test 1 · Paper 2

Question 10

What is the least possible value of $(xy-1)^2+(x+y)^2$ for real numbers x and y?

A 0

B $\frac{1}{4}$

C $\frac{1}{2}$

D 1

E 2

Full Mock Test 1 · Paper 2

Question 11

In the following list of numbers, the integer n appears n times in the list for $1 \leq n \leq 200$.

$1,\ 2,2,\ 3,3,3,\ 4,4,4,4,\ \ldots,\ 200,200,\ldots,200$

What is the median of the numbers in this list?

A 100.5

B 134

C 142

D 150.5

E 167

Full Mock Test 1 · Paper 2

Question 12

What is the sum of all real numbers x for which $|x^2 - 12x + 34| = 2$?

A 12

B 15

C 18

D 21

E 25

Full Mock Test 1 · Paper 2

Question 13

What is the value of $1 + 2 + 3 - 4 + 5 + 6 + 7 - 8 + \cdots + 197 + 198 + 199 - 200$?

A $9,800$

B $9,900$

C $10,000$

D $10,100$

E $10,200$

Full Mock Test 1 · Paper 2

Question 14

Four distinct real numbers a, b, c, d are used to define four points $A = (a, b)$, $B = (b, c)$, $C = (c, d)$, $D = (d, a)$.
The quadrilateral $ABCD$ has all four sides the same length

A if and only if $(a - b)^2 = (c - d)^2$

B for no values of a, b, c, d

C if and only if $(a - d)^2 = (b - c)^2$

D if and only if $a - b + c - d = 0$

E if and only if $(a - c)^2 = (b - d)^2$

Full Mock Test 1 · Paper 2

Question 15

Tom runs a small reptile education booth at the weekend market. He keeps notes on 13 snakes: 4 are purple and 5 are happy. From observing their "math tricks," he knows that

• all of his happy snakes can add,
• none of his purple snakes can subtract, and
• any snake that can't subtract also can't add.

Which of these conclusions must be true?

A Being purple is a necessary condition for being happy.

B Being able to subtract is a sufficient condition for being happy.

C Being able to add is a sufficient condition for being happy.

D Not being purple is a necessary condition for being happy.

E Not being able to subtract is a necessary condition for being happy.

Full Mock Test 1 · Paper 2

Question 16

Two transformations are said to commute if applying the first followed by the second gives the same result as applying the second followed by the first.

Label the transformations as T_1, T_2, T_3:

• T_1: a translation 2 units to the right;
• T_2: a $90°$-rotation counterclockwise about the origin;
• T_3: a reflection across the x-axis.

Consider these 3 distinct pairs of transformations: $(T_1, T_2),\ (T_1, T_3),\ (T_2, T_3)$. Of these pairs, how many commute? *("commute" means the order of the two transformations doesn't matter.)*

A 1

B 2

C 3

D None

Full Mock Test 1 · Paper 2

Question 17

Positive real numbers a and b have the property that $\sqrt{\log a}+\sqrt{\log b}+\log\sqrt{a}+\log\sqrt{b}=100$ and all four terms on the left are positive integers, where $\log$ denotes the base 10 logarithm. What is ab?

A 10^{52}

B 10^{100}

C 10^{144}

D 10^{164}

E 10^{200}

Full Mock Test 1 · Paper 2

Question 18

The equation $x^2 - 4kx + y^2 - 4y + 8 = k^3 - k$ is the equation of a circle

A if and only if either $-4 < k < -1$ or $k > 1$

B if and only if $k < -1$

C for all real values of k

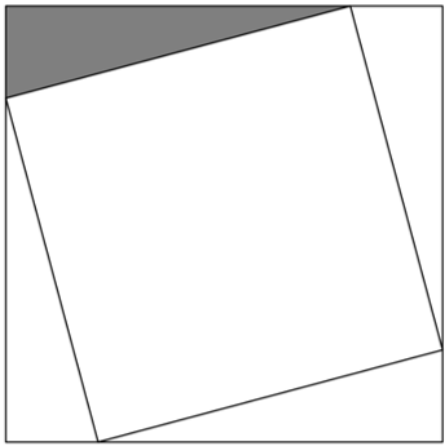
D if and only if $k > 1$

E if and only if either $-1 < k < 0$ or $k > 1$

Full Mock Test 1 · Paper 2

Question 19

A square of area 2 is inscribed in a square of area 3, creating four congruent triangles, as shown. What is the ratio of the shorter leg to the longer leg in the shaded right triangle?



A $\frac{1}{5}$

B $\frac{1}{4}$

C $2-\sqrt{3}$

D $\sqrt{3}-\sqrt{2}$

E $\sqrt{2}-1$

Full Mock Test 1 · Paper 2

Question 20

Define $P(x) = (x - 1^2)(x - 2^2) \cdots (x - 100^2)$. How many integers n are there such that $P(n) \leq 0$?

A 4900

B 4950

C 5000

D 5050

E 5100